

Multiple Choice (10 marks)

1. ^{23}Na has an NMR frequency of 198.52 MHz in a 750 MHz NMR spectrometer. What is its magnetogyric ratio?
 - (a) $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
 - (b) $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
 - (c) $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
 - (d) $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
2. The +1 oxidation state is more stable than the +3 oxidation state for which group 13 element?
 - (a) B
 - (b) Al
 - (c) In
 - (d) Tl
3. Cobalt-60 is used in the radiation therapy of cancer and can be produced by bombardment of cobalt-59 with which of the following?
 - (a) Neutrons
 - (b) Alpha particles
 - (c) Beta particles
 - (d) X-rays
4. Which of the following is always true of a spontaneous process?
 - (a) The process is exothermic.
 - (b) The process does not involve any work.
 - (c) The entropy of the system increases.
 - (d) The total entropy of the system plus surroundings increases.
5. Calculate the polarization of a proton in a magnetic field of 335 mT and 10.5 T at 298 K.
 - (a) 6.345×10^{-4} at 0.335 T; 9.871×10^{-5} at 10.5 T
 - (a) 0.793×10^{-4} at 0.335 T; 6.931×10^{-7} at 10.5 T
 - (a) 1.148×10^{-6} at 0.335 T; 3.598×10^{-5} at 10.5 T
 - (a) 4.126×10^{-3} at 0.335 T; 2.142×10^{-6} at 10.5 T

True / False (10 marks)

Answer True or False

6. True or False: Zn^{2+} is commonly used as a paramagnetic quencher because it has unpaired electrons in its d-orbitals.
7. True or False: Potassium hydrogen phthalate is a primary standard commonly used for standardizing bases in titrations.
8. True or False: The Larmor frequency for a proton in a magnetic field of 1 T is 42.58 MHz.
9. True or False: MgO has the greatest lattice enthalpy among the listed ionic substances due to its high charge density and ionic bond strength.
10. True or False: At standard temperature and pressure, bromine (Br_2) is a red-brown volatile liquid.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the ^{13}C NMR spectrum of hexane, explaining why it exhibits three distinct chemical shifts and how molecular structure influences these shifts.
12. Discuss the ^{13}C NMR spectrum of 2-methylpentane, explaining why it exhibits five distinct chemical shifts and how this relates to its molecular structure and isomerism.

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